

MARITIME LOGISTICS INTELLIGENCE

GLOBAL SUPPLY CHAIN DISRUPTION

2023–2025 Red Sea Maritime Crisis Analysis

SQL-Based Analysis of Route Diversion, Freight Rates, Port Delays, Trade Volume,
Trade Volume, Emissions & Supply Chain Vulnerability





GLOBAL MARITIME ROUTES

2023–2025 RED SEA DISRUPTION ANALYSIS

Key Corridors Impacted by Red Sea Disruption

ROUTE LEGEND

- Asia–Europe via Suez (Traditional Route)
- - - Asia–Europe via Cape of Good Hope (Alternative Route)
- - - Middle East–Europe Oil Route (Energy Corridor)
- Mediterranean Feeder (Regional / Short Sea)
- ⦿ Major Port
- Key Transshipment / Hub Port
- Red Sea Disruption Area

KEY PORTS IN DATASET

- Port Said (Egypt)
- Jeddah (Saudi Arabia)
- Dubai / Jebel Ali (UAE)
- Rotterdam (Netherlands)
- Hamburg (Germany)
- Antwerp (Belgium)
- Piraeus (Greece)
- Algiers (Algeria)
- Novorossiysk (Russia)
- Ceyhan (Turkey)
- Basra (Iraq)
- Singapore (Singapore)
- Shanghai (China)
- Ningbo (China)
- Shenzhen (China)
- Cape Town (South Africa)
- Aden (Yemen)



ROUTE SUMMARY COMPARISON

Route	Approx. Distance* (nm)	Key Characteristics	Status (2023–2025)
— Asia–Europe via Suez	~8,500	Shortest route via Suez Canal & Red Sea	Disrupted / High Risk
- - - Asia–Europe via Cape of Good Hope	~11,800	Longer but safer alternative around Africa	Active Alternative
- - - Middle East–Europe Oil Route	~6,200	Energy shipments from Gulf to Europe	Partially Disrupted
•••• Mediterranean Feeder	~1,500	Short-sea regional connections	Operational

*Distances are approximate and based on standard_distance_nm from dataset.



KEY INSIGHTS

- ⌚ Disruption in the Red Sea forces rerouting around Africa, adding distance, time & cost.
- 💰 Suez-linked routes show highest exposure to delay, cost pressure & emissions.
- 🌿 Longer routes via Cape of Good Hope increase fuel consumption & GHG emissions.
- 🛡️ Supply chain resilience depends on route diversification & real-time risk monitoring.

THE BUSINESS PROBLEM

WHY THE RED SEA DISRUPTION MATTERED

“Rerouting away from the Suez Canal changed the economics and operations of global maritime shipping.”

ROUTE COMPARISON

ASIA → SUEZ CANAL → EUROPE

Shorter, standard corridor — the pre-crisis default

ASIA → CAPE OF GOOD HOPE → EUROPE

Longer diversion around Africa — the crisis alternative

UNCTAD CONTEXT · Suez transits fell 40%+ from peak by Jan 2024 · Container tonnage via Suez fell 82% by early Feb 2024 · Cape of Good Hope tonnage rose 60% Dec 2023–Feb 2024

FOUR CASCADING IMPACTS

LONGER ROUTES → HIGHER OPERATIONAL EXPOSURE

FREIGHT PRESSURE → HIGHER COST RISK

DELAYS → SUPPLY CHAIN UNCERTAINTY

REROUTING → HIGHER EMISSIONS

FROM OPERATIONAL DATA TO BUSINESS INSIGHT



SQL CAPABILITIES

JOIN · GROUP BY · AGGREGATION · CTE · CASE LOGIC · WINDOW FUNCTION / RANK

Five business questions translate operational disruption into measurable supply-chain risk.



ANALYTICAL ROADMAP

FIVE BUSINESS QUESTIONS

Each question translates operational disruption into a measurable supply-chain risk.

01

ROUTE DIVERSION DISTANCE

How much additional distance is created by avoiding the Suez Canal?

02

FREIGHT RATES BY VESSEL

How did freight rates vary across different vessel types and phases?

03

PORT TRANSIT DELAYS

Which ports experienced the highest transit delays during the crisis?

04

TRADE VOLUME & EMISSIONS

How did trade volume and GHG emissions change across disruption phases?

05

CORRIDOR VULNERABILITY

Which trade corridor appears the most operationally vulnerable overall?



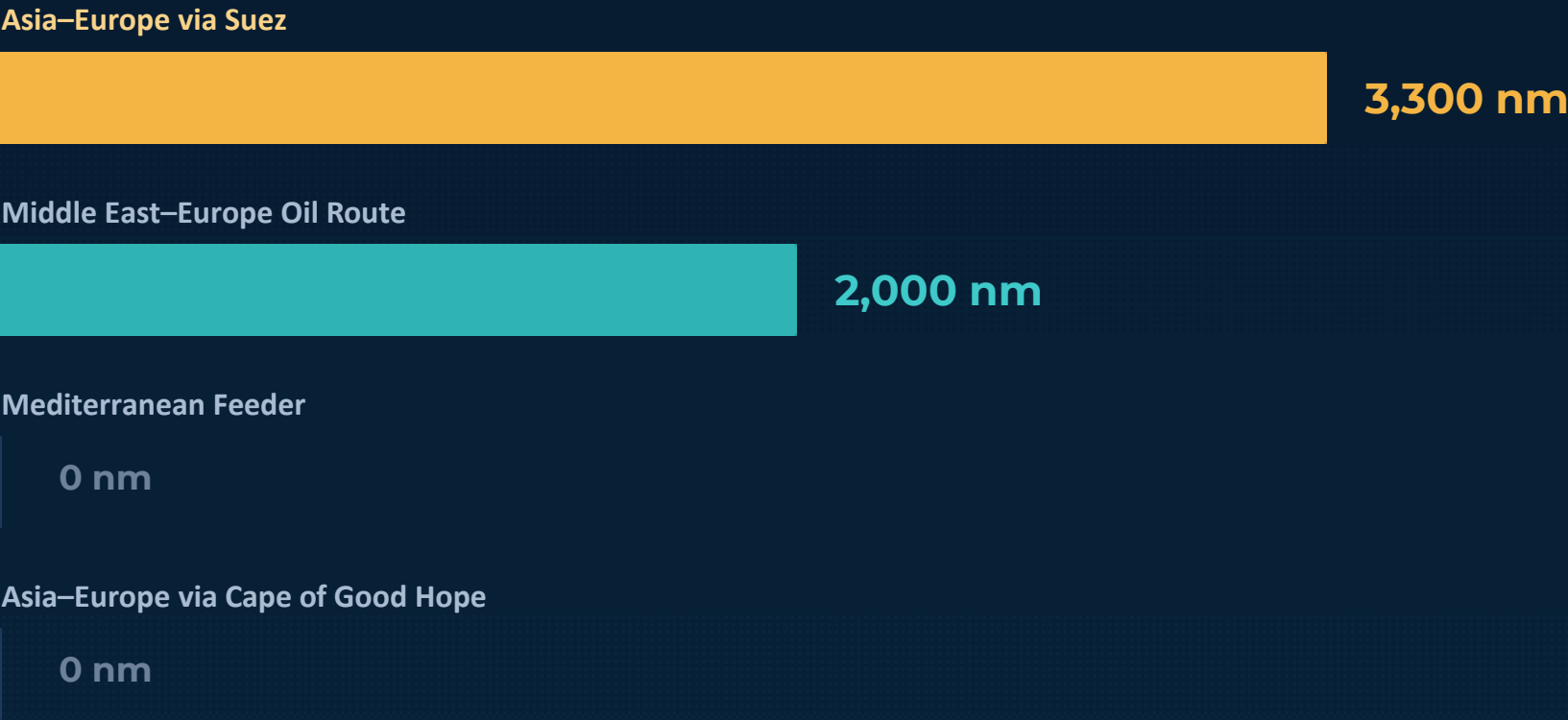
BUSINESS QUESTION 01

HOW MUCH ADDITIONAL DISTANCE IS CREATED BY AVOIDING SUEZ?

SQL

```
SELECT
  r.route_name,
  r.standard_distance_nm,
  r.diverted_distance_nm,
  (r.diverted_distance_nm - r.standard_distance_nm)
AS extra_distance_nm,
  ROUND(AVG(c.distance_nm), 2) AS
avg_actual_distance_nm,
  COUNT(c.record_id) AS total_trips
FROM crisis_metrics c
JOIN routes_master r
  ON c.route_id = r.route_id
GROUP BY
  r.route_name,
  r.standard_distance_nm,
  r.diverted_distance_nm
ORDER BY extra_distance_nm DESC;
```

Extra Distance Created by Route Diversion



KPI • LARGEST DIVERSION GAP

+3,300 NM

INSIGHT

Asia-Europe via Suez has the largest modeled diversion gap at 3,300 nautical miles. Longer rerouting increases sailing distance and operational exposure.

HOW DID FREIGHT RATES VARY ACROSS VESSEL TYPES?

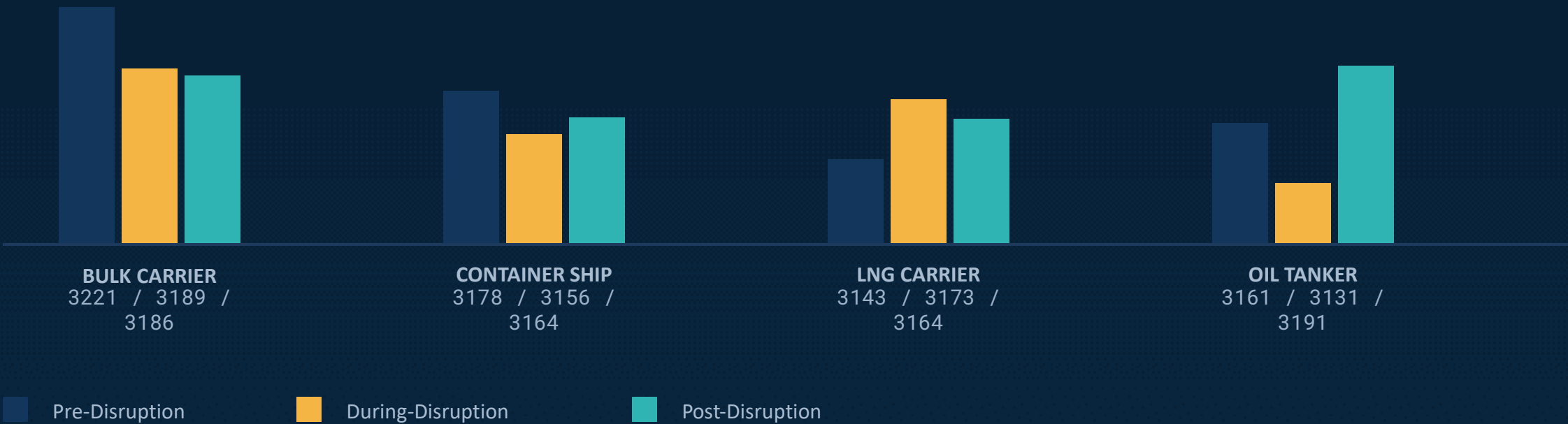
SQL

```
SELECT
  v.vessel_type,
  c.period_status,
  COUNT(c.record_id) AS total_shipments,
  ROUND(AVG(c.freight_rate_usd), 2) AS
avg_freight_rate_usd,
  ROUND(MAX(c.freight_rate_usd), 2) AS
max_freight_rate_usd,
  ROUND(MIN(c.freight_rate_usd), 2) AS
min_freight_rate_usd
FROM crisis_metrics c
JOIN vessels_master v
  ON c.vessel_type_id = v.vessel_type_id
GROUP BY
  v.vessel_type,
  c.period_status
ORDER BY
  v.vessel_type,
  c.period_status;
```

INSIGHT

Freight-rate movements were not uniform across vessel types. Oil tankers peaked post-disruption; LNG carriers peaked during the disruption phase.

Average Freight Rate (USD) by Vessel Type & Phase



WHICH PORTS EXPERIENCED THE HIGHEST TRANSIT DELAYS?

SQL

```
SELECT
  p.port_name,
  p.country,
  p.base_dwell_days,
  ROUND(AVG(c.transit_delay_days), 2) AS
avg_transit_delay_days,
  ROUND(MAX(c.transit_delay_days), 2) AS
max_transit_delay_days,
  COUNT(c.record_id) AS total_port_calls
FROM crisis_metrics c
JOIN ports_congestion_master p
  ON c.port_id = p.port_id
GROUP BY
  p.port_name,
  p.country,
  p.base_dwell_days
ORDER BY avg_transit_delay_days DESC;
```

KPI • AVERAGE PORT DELAY

~8 DAYS

Average Transit Delay (Days)



INSIGHT

All four ports show approximately eight days average delay, with Jeddah highest at 8.06 days — 8.06 days — indicating significant operational pressure across maritime hubs.

HOW DID TRADE VOLUME AND GHG EMISSIONS CHANGE ACROSS DISRUPTION PHASES?

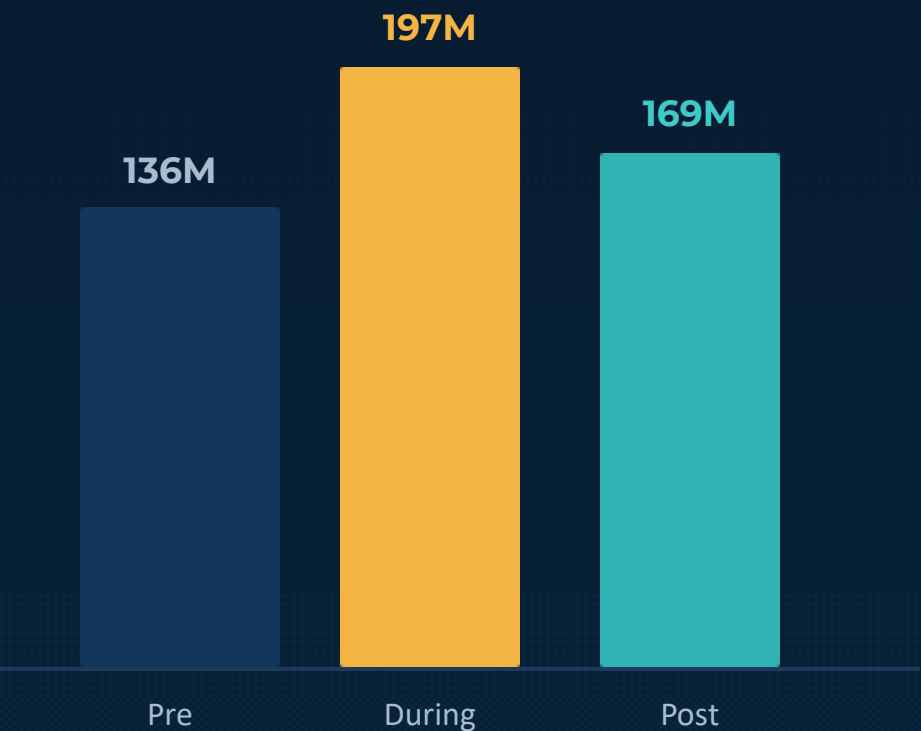
SQL

```
SELECT
  c.period_status,
  COUNT(c.record_id) AS
total_shipments,
  ROUND(SUM(c.trade_volume_tons),
2) AS total_trade_volume_tons,
  ROUND(AVG(c.trade_volume_tons),
2) AS avg_trade_volume_per_shipment,
  ROUND(SUM(c.ghg_emission_tons),
2) AS total_ghg_emission_tons
FROM crisis_metrics c
GROUP BY c.period_status
ORDER BY total_trade_volume_tons
DESC;
```

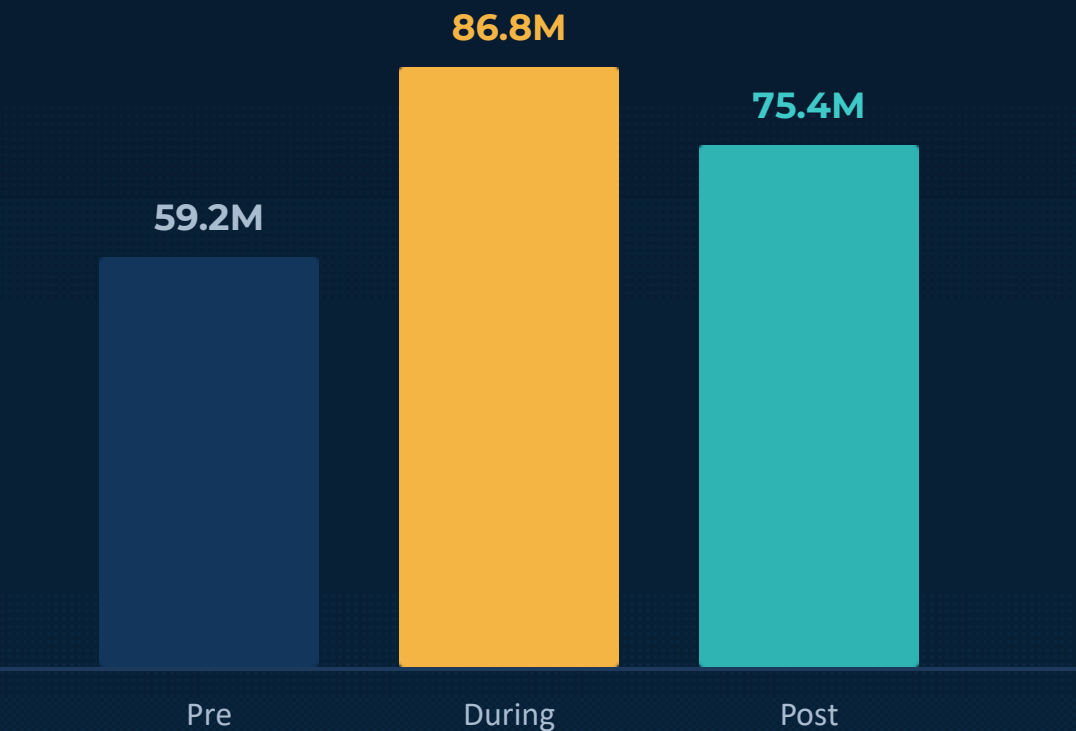
KPI • PEAK EMISSIONS

86.8M TONS
GHG

Total Trade Volume (million tons)



Total GHG Emissions (tons)



Exact GHG values — Pre: 59,205,286 · During: 86,784,893 · Post: 75,362,151 tons. Note: project dataset analytical results, not UNCTAD raw shipment data.

INSIGHT

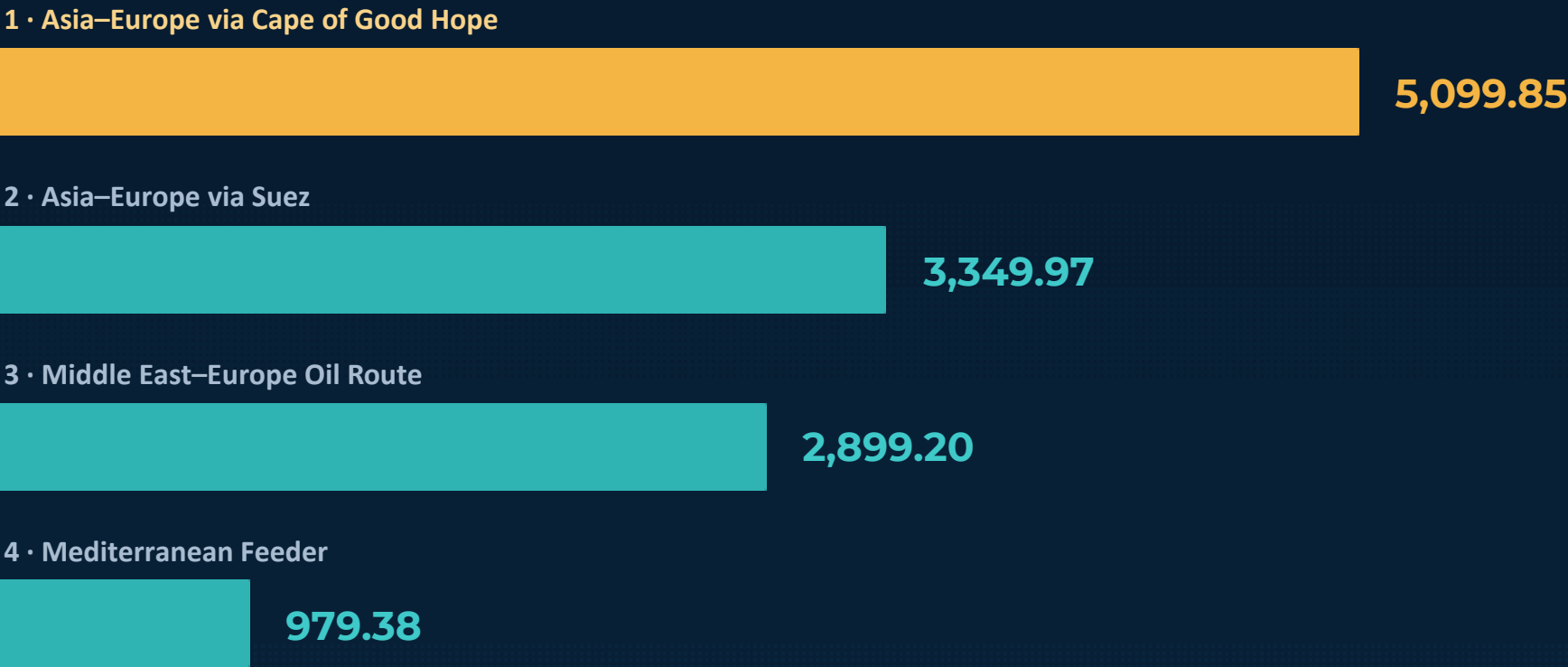
The During-Disruption phase records the highest modeled trade volume and GHG emissions. Longer rerouting increases fuel use and emissions.

WHICH TRADE CORRIDOR APPEARS MOST OPERATIONALLY VULNERABLE?

SQL

```
WITH RouteMetrics AS (  
  SELECT  
    r.route_name,  
    COUNT(c.record_id) AS total_shipments,  
    ROUND(AVG(c.distance_nm), 2) AS avg_distance,  
    ROUND(AVG(c.transit_delay_days), 2) AS  
avg_delay,  
    ROUND(AVG(c.freight_rate_usd), 2) AS  
avg_freight_rate,  
    ROUND(  
      (AVG(c.distance_nm) * 0.3)  
      + (AVG(c.transit_delay_days) * 50)  
      + (AVG(c.freight_rate_usd) * 0.2),  
      2  
    ) AS composite_risk_score  
  FROM crisis_metrics c  
  JOIN routes_master r  
    ON c.route_id = r.route_id  
  GROUP BY r.route_name  
)  
SELECT  
  route_name,  
  total_shipments,  
  avg_distance,  
  avg_delay,  
  avg_freight_rate,  
  composite_risk_score,  
  RANK() OVER (  
    ORDER BY composite_risk_score DESC  
  ) AS risk_rank  
FROM RouteMetrics;
```

Composite Vulnerability Score (Ranked)



INSIGHT

Cape of Good Hope ranks highest — combining the greatest distance, delay, and freight rate into analytical climax of corridor vulnerability.

FORMULA

Composite Score = (Avg Distance × 0.3) + (Avg Delay × 50) + (Avg Freight Rate × 0.2)

Methodology: weights and scaling are project-specific analytical assumptions.

WHAT DOES THE ANALYSIS TELL US?

ROUTE

+3,300 NM

Suez-dependent Asia–
Asia–Europe routing
carries the largest
modeled diversion
burden.

COST

**NON-UNIFORM
FREIGHT
RESPONSE**

Freight-rate patterns
differ by vessel type
and disruption phase.

DELAY

~8 DAYS

All four analyzed ports
show ~8 days
average transit delay.
delay.

ENVIRONMENT

**86.8M TONS
GHG**

During-Disruption
phase records the
highest aggregate
modeled GHG
emissions.

RISK

5,099.85

Cape of Good Hope
corridor ranks highest
highest on the
composite vulnerability
vulnerability score.

ROUTE DISRUPTION → COST & DELAY PRESSURE → ENVIRONMENTAL IMPACT → CORRIDOR VULNERABILITY

FINAL RECOMMENDATIONS

FROM SQL FINDINGS TO BUSINESS ACTION

How can supply-chain operators respond to the risks identified? PROBLEM → FINDING → RECOMMENDATION

PROBLEM · Large route diversion burden	FINDING · +3,300 nm Asia–Europe via Suez	RECOMMENDATION · Develop pre-planned alternative alternative routing and contingency corridors.
PROBLEM · Freight exposure varies by vessel type	FINDING · Freight-rate patterns patterns vary by vessel and phase	RECOMMENDATION · Monitor vessel-specific freight exposure exposure and prepare contingency contracts.
PROBLEM · High transit delays	FINDING · ~8 days average delay	RECOMMENDATION · Pre-plan alternative ports and surge surge capacity.
PROBLEM · Higher emissions during disruption	FINDING · 86.8M tons during disruption phase	RECOMMENDATION · Include emissions in route and speed speed decisions.
PROBLEM · Corridor vulnerability concentration	FINDING · Cape of Good Hope scores highest	RECOMMENDATION · Prioritise corridor monitoring, scenario scenario planning, contingency routing.

“Resilient supply chains require more than alternative routes—they require continuous visibility into distance, cost, delay and environmental exposure.”

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THANK YOU

Questions & Discussion

Global Supply Chain Disruption · 2023–2025 Red Sea Maritime Crisis Analysis



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